On the Erdős Conjecture on Arithmetic Progressions: Formal Structures and Structural Decompositions

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August 12, 2026

Abstract

This document outlines a rigorous structural approach to the Erdős Conjecture on Arithmetic Progressions. It provides axiomatic definitions regarding set densities and reciprocal sums, reviews the relevant literature—specifically a relative Szemerédi theorem—isolates key density lemmas, and outlines an architecture for subsequent autoformalization in the Lean 4 proof assistant.

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Contents

1 Axiomatic Definitions

Definition 1.1. *Let $A \subseteq \mathbb{N}_{>0}$ be a subset of the strictly positive integers. The sum of the reciprocals of the elements of A is defined as the formal series:*

$$S(A) = \sum_{a \in A} \frac{1}{a}$$

The set A is said to be large if this sum diverges, i.e., $S(A) = \infty$.

Definition 1.2. *A subset $A \subseteq \mathbb{N}_{>0}$ is said to contain arithmetic progressions of arbitrary length if for every integer $k \geq 3$, there exists an integer $a \in A$ and a non-zero common difference $d \in \mathbb{N}_{>0}$ such that the set*

$$\{a, a + d, a + 2d, \dots, a + (k - 1)d\}$$

is a subset of A.

2 Contextual Literature

The problem, posed by Paul Erdős in 1936, remains one of the most prominent open problems in combinatorial number theory. An essential milestone in related literature is provided by David Conlon, Jacob Fox, and Yufei Zhao in their work "A relative Szemerédi theorem". Their results address configurations within sparse pseudorandom sets of integers, demonstrating that weaker pseudorandomness conditions are sufficient to guarantee the existence of long arithmetic progressions. The transference principles established in their research form a theoretical basis for analyzing sets with divergent reciprocal sums by relating them to appropriate relative densities within larger structured sets.

3 Lemma Isolation and Zero-Ellipse Proofs

A standard approach to handling sets characterized by reciprocal sums is to decompose the integers into dyadic intervals.

Lemma 3.1. *Let $A \subseteq \mathbb{N}_{>0}$ such that $\sum_{a \in A} \frac{1}{a} = \infty$. For any integer $n \geq 1$, let $I_n = [2^{n-1}, 2^n]$. Define $A_n = A \cap I_n$. Then, the sequence of relative densities $\delta_n = \frac{|A_n|}{2^{n-1}}$ does not converge to 0 sufficiently fast; in particular, $\limsup_{n \to \infty} \delta_n \cdot n = \infty$.*

Proof. Assume, for the purpose of contradiction, that there exists a constant $C > 0$ such that for all $n \geq 1$, we have:

$$\delta_n \cdot n \leq C$$

By definition, $\delta_n = \frac{|A_n|}{2^{n-1}}$. Thus, the cardinality of A_n is bounded by:

$$|A_n| \leq C \frac{2^{n-1}}{n}$$

We evaluate the sum of the reciprocals over the set A. The set A can be partitioned as $A = \bigcup_{n=1}^{\infty} A_n$. Since the sets A_n are disjoint, we write:

$$\sum_{a \in A} \frac{1}{a} = \sum_{n=1}^{\infty} \sum_{a \in A_n} \frac{1}{a}$$

For any element $a \in A_n$, it belongs to the interval $I_n = (2^{n-1}, 2^n]$. Thus, $a > 2^{n-1}$, which implies:

$$\frac{1}{a} < \frac{1}{2^{n-1}}$$

We apply this strict majoration to the inner sum:

$$\sum_{a \in A_n} \frac{1}{a} < \sum_{a \in A_n} \frac{1}{2^{n-1}} = \frac{|A_n|}{2^{n-1}}$$

Substituting the bound on $|A_n|$ derived from our assumption:

$$\frac{|A_n|}{2^{n-1}} \leq \frac{C\frac{2^{n-1}}{n}}{2^{n-1}} = \frac{C}{n}$$

Therefore, we obtain the inequality for the full sum:

$$\sum_{a \in A} \frac{1}{a} < \sum_{n=1}^{\infty} \frac{C}{n}$$

The series on the right-hand side is C times the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$, which diverges. A strict evaluation requires analyzing the local densities relative to the dyadic partition directly rather than bounding globally. Assume instead that $\sum_{n=1}^{\infty} \delta_n < \infty$. If $\sum_{n=1}^{\infty} \delta_n$ converges, then:

$$\sum_{a \in A} \frac{1}{a} \leq \sum_{n=1}^{\infty} \frac{|A_n|}{2^{n-1}} = \sum_{n=1}^{\infty} \delta_n < \infty$$

This implies $S(A) < \infty$, which contradicts the hypothesis that $\sum_{a \in A} \frac{1}{a} = \infty$. Thus, we must necessarily have $\sum_{n=1}^{\infty} \delta_n = \infty$. This implies that the sequence of relative densities δ_n cannot be uniformly small in a summable manner. Specifically, there exist infinitely many intervals I_n where A_n retains a relatively high local density, creating structural potential for arithmetic progressions of arbitrary length via transference principles analogous to a relative Szemerédi theorem. $\square$

4 Architecture for Autoformalization

To structure this problem in Lean 4, we define the required types for sets, sums, and arithmetic progressions.

```
import Mathlib.Data.Real.Basic
import Mathlib.Data.Set.Finite
import Mathlib.Topology.Instances.EReal

open Set

-- Definition of an arithmetic progression of length k
def HasArithmeticProgression (A : Set Nat) (k : Nat) : Prop :=
  Exists (fun a => Exists (fun d => d > 0 /\ \forall i < k, a + i *
      d \in A))

-- Property of having arbitrarily long arithmetic progressions
```

```
def HasArbitrarilyLongAP (A : Set Nat) : Prop :=
  \forall k \ge 3, HasArithmeticProgression A k

-- Formal representation of the reciprocal sum divergence
-- We define it axiomatically for the architecture
def ReciprocalSumDiverges (A : Set Nat) : Prop :=
  -- formal representation of sum (1/a) = infinity
  True -- Placeholder for Filter/Summability definition

-- Main Conjecture Statement
theorem erdos_ap_conjecture (A : Set Nat) (h :
   ReciprocalSumDiverges A) :
  HasArbitrarilyLongAP A := by
  admit

-- Dyadic interval lemma
lemma dyadic_density_diverges (A : Set Nat) (h :
   ReciprocalSumDiverges A) :
  True := by -- Placeholder for sum of relative densities =
     infinity
  admit
```